

Kourtesis Dimitris - Data Analysis - Set 5

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2 E

1:

Θ $\mathbf{X} = (X_1, X_2, \dots, X_n),$ Π 10^5 /

Γ :

$$M = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - M)^2$$

$$S^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - M)^2$$

Σ $n (\cdot \cdot \cdot n = 5, 10, \dots)$ σ^2 $\cdot \Sigma$,

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: # -----
# P
# -----
N_exp = 10**5
n_values = [5, 10, 20, 50, 100, 500]

mu = 0
sigma = 1

rng = np.random.default_rng(seed=123)

results = []
```

```

# -----
#  $\Delta$           3 x 6
# -----
fig, axes = plt.subplots(
    nrows=3,
    ncols=len(n_values),
    figsize=(24, 10)
)

for col, n in enumerate(n_values):

    #  $T$            $N(0,1)$ 
    X = rng.normal(loc=mu, scale=sigma, size=(N_exp, n))

    #  $E$ 
    M = np.mean(X, axis=1)

    S2 = np.mean((X - M[:, None])**2, axis=1)

    S2_star = np.sum((X - M[:, None])**2, axis=1) / (n - 1)

    #  $A$ 
    results.append({
        "n": n,
        "E[M]": np.mean(M),
        "Var(M)": np.var(M),
        "E[S2]" : np.mean(S2),
        " $\theta$  E[S2]" : (n - 1) / n,
        "E[S*2]" : np.mean(S2_star),
        " $\theta$  E[S*2]" : 1
    })

# -----
# 1          :  $M$ 
# -----
ax = axes[0, col]
ax.hist(M, bins=80, density=True, alpha=0.75)
ax.axvline(mu, linestyle="--")
ax.set_title(rf"$n={n}$")
ax.set_xlabel(r"$M$")
if col == 0:
    ax.set_ylabel(r"$K$")

# -----
# 2          :  $S^2$ 
# -----
ax = axes[1, col]

```

```

ax.hist(S2, bins=80, density=True, alpha=0.75)
ax.axvline(sigma**2, linestyle="--")
ax.set_xlabel(r"$S^2$")
if col == 0:
    ax.set_ylabel(r"$K \quad S^2$")

# -----
# 3 : S*^2
# -----
ax = axes[2, col]
ax.hist(S2_star, bins=80, density=True, alpha=0.75)
ax.axvline(sigma**2, linestyle="--")
ax.set_xlabel(r"$S^{\{2\}}$")
if col == 0:
    ax.set_ylabel(r"$K \quad S^{\{2\}}$")

fig.suptitle(
    r"$K \quad M$, $S^2 \quad S^{\{2\}}$",
    fontsize=18
)

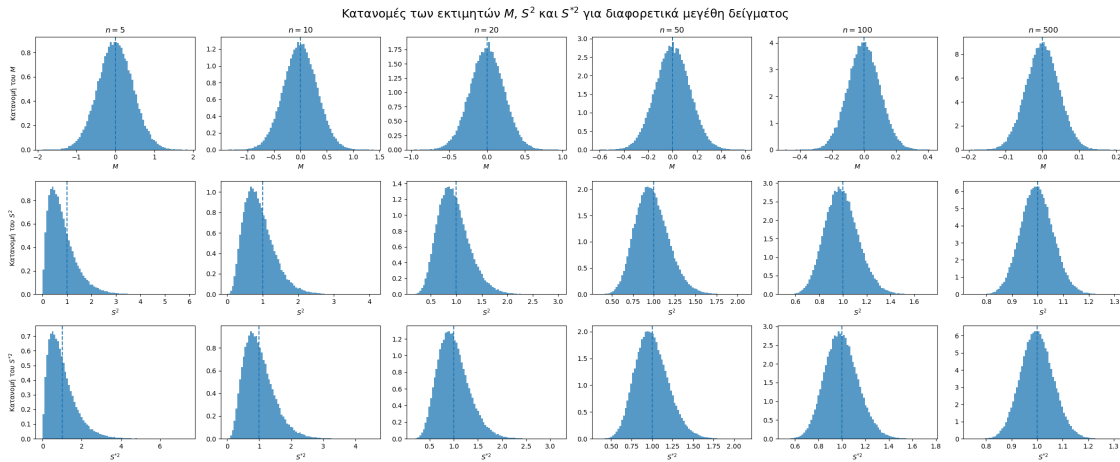
plt.tight_layout()
plt.savefig("estimators_histograms_3x6.png", dpi=300)
plt.show()

# -----
# II
# -----
df = pd.DataFrame(results)

# I
df_rounded = df.round(5)

print(df_rounded)

```



	n	E[M]	Var(M)	E[S ²]	θ	E[S ²]	\
0	5		0.00015	0.19927		0.80009	0.800
1	10		0.00126	0.09958		0.89998	0.900
2	20		0.00021	0.04941		0.94906	0.950
3	50		0.00017	0.01992		0.97902	0.980
4	100		-0.00016	0.00995		0.98983	0.990
5	500		0.00019	0.00200		0.99779	0.998

	E[S* ²]	θ	E[S* ²]
0		1.00011	1
1		0.99998	1
2		0.99901	1
3		0.99900	1
4		0.99983	1
5		0.99979	1

2.1 Σ

Ο

$$X \sim N(0, 1),$$

$$\mu = 0, \quad \sigma^2 = 1.$$

2.1.1 Ε M

Ο

$$M = \frac{1}{n} \sum_{i=1}^n X_i$$

A μ M 0 , n . A

$$E[M] = \mu = 0.$$

A M E , n , M 0 . A

$$\text{Var}(M) = \frac{\sigma^2}{n}.$$

E, $n \rightarrow \infty$,

$$\text{Var}(M) \rightarrow 0.$$

A M μ .

2.1.2 E S^2
O

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - M)^2$$

A S^2

$$\sigma^2 = 1.$$

Θ

$$E[S^2] = \frac{n-1}{n} \sigma^2.$$

A S^2 , σ^2 .
 Ω , n ,

$$\frac{n-1}{n} \rightarrow 1.$$

E S^2 . A
1. A , , S^2 .

2.1.3 E S^{*2}
O

$$S^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - M)^2$$

A

$$\sigma^2 = 1$$

$$n. \Theta$$

$$E[S^{*2}] = \sigma^2.$$

A S^{*2} .
E , , 1. E S^{*2} .

2.1.4 Σ

Σ :

- O M μ .
- O S^2 , σ^2 .
- O S^2 , n .
- O S^{*2} σ^2 .

2.2 Γ

M $n-1$ n ;
 n , n . Π , μ

$$\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$

Σ μ

$$M = \frac{1}{n} \sum_{i=1}^n X_i.$$

A : M « » . E ,

$$X_i - M$$

$$\mu.$$

2.2.1 M

A :

$$X_1, \quad X_2.$$

M

$$M = \frac{X_1 + X_2}{2},$$

. A ,

$$X_1 - M,$$

$$X_2 - M = -(X_1 - M).$$

Δ

M , « » .

2.2.2 B

Γ , n ,

$$\sum_{i=1}^n (X_i - M) = 0.$$

A $n - 1$, .

A $n - 1$.

Λ $n - 1$ (*degrees of freedom*).

2.2.3 Γ $n - 1$;

A n ,
 M .

H $n - 1$.

M :

- $1/n$ n ,
 - $1/(n - 1)$,
 - .
-

2.2.4 Σ

O $n - 1$ « » . Π :

1. Δ μ .
2. T .
3. H .
4. Γ , $n - 1$ n .

A **Bessel (Bessel’s correction)**

$$S^{*2} = \frac{1}{n - 1} \sum_{i=1}^n (X_i - M)^2$$

,

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - M)^2$$

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